

⇒ Taylor's series:-

The Taylor series is a way to break a complicated function into a sum of simpler pieces (polynomials) that are easy to work with. it helps us approximate the value of a function near a specific point by adding terms based on its derivatives. The more terms we add, the more ~~less~~ accurate the approximation.

→ formula:-

$$y_1 = y(n) = y_0 + (n-n_0)y'_0 + \frac{(n-n_0)^2}{2!}y''_0 + \dots$$

Let $(n-n_0) = h$. it is the length of the interval.

$$y_1 = y_0 + hy'_0 + \frac{h^2}{2!}y''_0 + \frac{h^3}{3!}y'''_0 + \dots$$

$$y_2 = y_1 + hy'_1 + \frac{h^2}{2!}y''_1 + \frac{h^3}{3!}y'''_1 + \dots$$

Q1. ex

Use Taylor's series method to solve
 $\frac{dy}{dx} = 2y + 3e^x$ with the initial condition

$$x_0 = 0, y_0 = 1$$

find approximate value of y
 $x_1 = 0.1$ and $x_2 = 0.2$

Formula, $y(x_1) = y_0 + h_1 y' + \frac{h_1^2 y''}{2!} + \frac{h_1^3 y'''}{3!} + \dots$

$$h = (x_1 - x_0) = 0.1 - 0 = 0.1$$

$$y(0.1) = y_0 + h_1 y' + \frac{h_1^2 y''}{2!} + \frac{h_1^3 y'''}{3!} + \dots$$

$$y' = 2y + 3e^x$$

$$y = 2(0) + 3e^0 = 3$$

$$y' = 2y + 3e^{x_0}$$

$$y = 2(3) + 3e^0 = 9$$

$$y'' = 2y' + 3e^{x_0}$$

$$y = 2(9) + 3e^0 = 21$$

$$y''' = 2y'' + 3e^{x_0}$$

$$y = 2(21) + 3e^0 = 45$$

$$y(0.1) = 1 + (0.1)(3) + \frac{(0.1)^2(9)}{2!} + \frac{(0.1)^3(21)}{3!} + \frac{(0.1)^4(45)}{4!} + \dots$$

$$= 1.570$$

→ Now Given

$$n_0 = 0 \quad y_0 = 1$$

Find the approx val of y

$$n_1 = 0.1 \quad \text{and} \quad n_2 = 0.2$$

$$n_2 - n_1 \quad 0.2 - 0.1 = \boxed{0.1} \rightarrow h$$

$$y_1 = 1.570, \quad n_2 = 0.2$$

y_1

$$y' = 2y_1 + 3e^n \rightarrow y' = 2(1.570) + 3e^{0.1} \\ = 6.4555$$

$$y'' = 2y' + 3e^n \rightarrow y'' = 2(6.4555) + 3e^{0.1} \\ = 16.2265$$

$$y''' = 2y'' + 3e^n \rightarrow y''' = 2(16.2265) + 3e^{0.1} \\ = 35.7685$$

$$y^{(4)} = 2(35.7685) + 3e^{0.1} = \boxed{74.8525}$$

By Taylor's series $y_2 = y(n_2) = y_1 + hy'_1 + \frac{h^2}{2!}y''_1 + \frac{h^3}{3!}y'''_1 + \dots$

$$y(0.2) = 1.570 + (0.1)(6.4555) + \frac{(0.1)^2}{2!}(16.2265) + \frac{(0.1)^3}{3!}(35.7685) + \frac{(0.1)^4}{4!}(74.8525) + \dots$$

$$2.3029 = 2.303$$

upto 3d

Ans

→ class work

Q3 Solve $\frac{dy}{dx} = x+y = f(x,y)$
sol

Formula: $y = y_0 + hy' + \frac{h^2 y''}{2!} + \frac{h^3 y'''}{3!} + \frac{h^4 y''''}{4!} + \dots$

Given

$x=1$, $y=0$ and carry to 1.2
with $h=0.1$

$x_0=1$, $y_0=0$, $h=0.1$

$x_1 = x_0 + h = 1 + 0.1 = 1.1$

$y(1.1) = ??$

$y' = x+y \rightarrow 1+0=1$

$y'' = 1+y' \rightarrow 1+1=2$

$y''' = y'' \rightarrow 2$

$y'''' = y''' \rightarrow 2$

$y(1.1) = 0 + (0.1)(1) + \frac{(0.1)^2(2)}{2!} + \frac{(0.1)^3(2)}{3!} + \frac{(0.1)^4(2)}{4!}$

$= 0.1103$ upto 4 decimal
 y_1 row for y_2

$y_2 = y_1 + hy'_1 + \frac{h^2 y''_1}{2!} + \frac{h^3 y'''_1}{3!} + \frac{h^4 y''''_1}{4!} + \dots$

$$n_2 = n_1 + h = 1.1 + 0.1 = \boxed{1.2}$$

$$y(1.2) = y_1 + hy'_1 + \frac{h^2 y''_1}{2!} + \frac{h^3 y'''_1}{3!} + \frac{h^4 y^{(4)}_1}{4!} + \dots$$

$$y_2(1.2) = 0.1103 + (0.1)(1.2103) + \frac{(0.1)^2}{2!}(2.2103) +$$

$$\frac{(0.1)^3}{3!}(2.2103) + \frac{(0.1)^4}{4!}(2.2103) \dots$$

$$= 0.24216 \text{ up to 4 decimal places}$$

Ans.

$$y'' = n + y = 1.1 + 0.1103 = \boxed{1.2103} \quad \uparrow$$

$$y''_2 = 1 + y' = 1 + 1.2103 = \boxed{2.2103}$$

$$y'''_3 = y'' = 2.2103 \rightarrow y''' \rightarrow \boxed{2.2103}$$